

**Programme: B E – Information Science & Engineering**
**Internal Assessment – II**

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|-------------------------------------------------------------|-------------------------------|
| TERM : 15 <sup>th</sup> April to 27 <sup>th</sup> July 2024 | COURSE NAME: Microcontrollers |
| DATE : 23/07/2024 TIME:9.30am-10.30pm                       | COURSE CODE : IS42            |
| MAX MARKS: 30                                               | PORTIONS : L21 - L42          |



Mobile Phones are banned

Instructions to Candidates: Answer any TWO full questions.

Marks: 15x2=30

| Q. NO | Questions                                                                                                                                                                                                                               | Blooms Levels (L1 to L6)* | CO  | Marks |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|-------|
| 1 a)  | Write a block diagram of a ARM7 Processor. List the modes of operation in ARM7.                                                                                                                                                         | L2                        | CO3 | 05    |
| b)    | Mention the steps of execution for the following instructions: STMDB R1!, {R7, R2-R4}, if R1=0x44400088.                                                                                                                                | L3                        | CO4 | 05    |
| c)    | List the features of cortex-M0. Mention the advantages for a designer to choose M0 for his design.                                                                                                                                      | L2                        | CO5 | 05    |
| 2 a)  | Explain ARM7 Assembler directives.                                                                                                                                                                                                      | L2                        | CO3 | 05    |
| b)    | What is the content of R6 after execution of following instructions? Given that R6=0x44001100 and R2=0x02<br>i. LDR R2, [R6, #0056]!<br>ii. STR R1, [R6, R2]!<br>iii. LDR R3, [R6]<br>Justify the results.                              | L3                        | CO4 | 05    |
| c)    | Explain the modes and states of Cortex-M0.                                                                                                                                                                                              | L2                        | CO5 | 05    |
| 3 a)  | Given R0=0x12345678, R1=0Xd5678007, R3=0x14<br>Find the contents of the destination registers, and the state of the relevant flags after execution of following instructions<br>i. MVN R7, R0<br>ii. MOV R5, R1<br>iii. ADDS R6, R0, R1 | L3                        | CO3 | 05    |
| b)    | Write a C program to display the string 'MICROCONTROLLERS' in the serial window of UART1.                                                                                                                                               | L3                        | CO4 | 05    |
| c)    | Discuss the interrupts of cortex-M0.                                                                                                                                                                                                    | L2                        | CO5 | 05    |

\*L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create